

Your grade: 71.42%

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Module 4 Summative Assessment

Try again

Review Learning Objectives

Abstract

2. In the context of smoothing splines, what is "overfitting"?
- ☐ Fitting a spline to a dataset with too many data points.
 - ☐ Using a spline with too few knots.
 - ☒ Fitting the spline too closely to the data, capturing noise rather than the underlying trend.
 - ☐ A situation where the smoothing spline is too rough.

Correct Overfitting occurs when a model captures noise instead of the true pattern.

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2. How does adding a penalty term based on the size of model coefficients affect a machine learning model?
- ☐ It removes redundant features from the dataset.
- ☐ To pass you need at least 80%. We keep your highest score.
- ☐ It increases the likelihood of model fitting noise in the data.
- ☒ It encourages the model to balance fitting the data with maintaining simpler model complexity.
- ☐ It leads to larger coefficients in the final model.

3. How does a cubic spline differ from a piecewise linear regression?
- ☐ A cubic spline is always a linear function.
 - ☒ Cubic splines use higher-degree polynomials than piecewise linear regression.
 - ☐ A cubic spline uses third-degree polynomials and ensures smoothness at the knots.
 - ☐ Cubic splines cannot fit non-linear data patterns.

Incorrect
Incorrect. While cubic splines use higher-degree polynomials, this is not their defining characteristic compared to piecewise linear regression.

4. Why are smoothing splines preferred over polynomial regression in some cases?
- ☒ They provide flexibility in fitting data and avoid problems like Runge's phenomenon in polynomial regression.
 - ☐ They are always computationally less expensive.
 - ☐ They can only be used with linear models.
 - ☐ They are simpler to implement than polynomial regression.
- ☒ **Correct**
Correct: Smoothing splines offer more flexibility and can avoid issues like oscillations at the boundaries.

5. What is the main purpose of adding a regularization term to a machine learning model's loss function?
- ☐ To increase the model's prediction accuracy on the training data.
 - ☐ To enhance the model's speed and computational efficiency.
 - ☒ To reduce overfitting by penalizing more complex models.
 - ☐ To automatically select the most relevant features in the dataset.
- ☒ **Correct**
Correct! Regularization aims to reduce overfitting by penalizing model complexity.

6. In the context of machine learning models, how does the use of a kernel function typically impact model performance?
- ☐ It directly reduces the training time of the model.
 - ☐ It allows models to capture more complex patterns by transforming data into a higher-dimensional space.
 - ☐ It simplifies the model to prevent overfitting.
 - ☒ It reduces the need for feature engineering.
- ☒ Incorrect
- Incorrect. Kernel functions transform data but do not eliminate the need for feature engineering.

7. In the context of piecewise polynomial regression, what is a "knot"?
- ☒ A point where the polynomial pieces connect or join together.
 - ☐ A point of maximum curvature in the polynomial model.
 - ☐ The highest degree of the polynomial used.
 - ☐ A parameter that controls the overall fit of the model.
- ☒ **Correct**
Correct! Knots are the points that separate different polynomial segments in piecewise regression.